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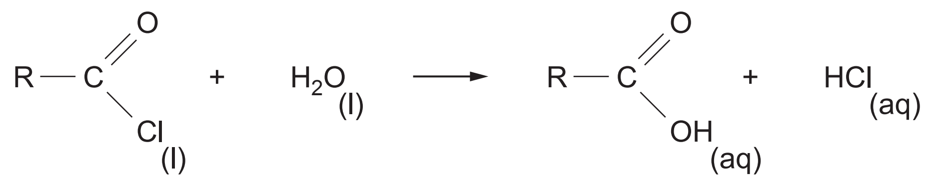
9. (a) 1.598 g of an acid chloride, RCOCl, was added to water.
The aqueous solution contained hydrochloric acid and a carboxylic acid, R — COOH, where R is an alkyl group



(i) State the type of reaction occurring. [1]

....

(ii) This acidic solution was titrated with 0.400 mol dm^{-3} aqueous sodium hydroxide, using a suitable indicator. Both acids were just neutralised by 75.00 cm^3 of the sodium hydroxide solution.

Use the results to calculate the relative molecular mass of the acid chloride. [3]

M_r ....

(iii) The low resolution ^{1}H NMR spectrum of the acid chloride showed two signals in the peak area ratio of 6:1.

Use this information and the relative molecular mass of the RCOCl, obtained in part (ii), to find the structure of the acid chloride. [2]

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(b) (i) The acid chloride, benzene-1,4-dicarbonyl dichloride, $ClOC — C_6H_4 — COCl$ is made by reacting benzene-1,4-dicarboxylic acid with phosphorus(V) chloride. The other products of this reaction are hydrogen chloride and phosphoryl trichloride, $POCl_3$.

Give the equation for this reaction. [1]

(ii) Benzene-1,4-dicarbonyl dichloride reacts with benzene-1,4-diamine to give a polyamide.

Show the repeating unit for this polyamide. [1]

(c) Nylon polyamides that are produced from starting materials with a different number of carbon atoms are given numbers.
For example Nylon 4,5 has a 4-carbon diamine fragment and a 5-carbon dicarboxylic acid fragment.

Butane-1,4-diamine can be used as a starting material for this polyamide. This can be produced in a two-stage reaction from butane-1,4-diol.

$$HOCH_2(CH_2)_2CH_2OH \xrightarrow[\text{Stage 1}]{\text{reagent } \mathbf{A}} ClCH_2(CH_2)_2CH_2Cl \xrightarrow[\text{Stage 2}]{\text{reagent } \mathbf{B}} H_2N(CH_2)_4NH_2$$

(i) State the name of reagent(s) **A**. [1]

....

(ii) State the name of reagent(s) **B**. [1]

....

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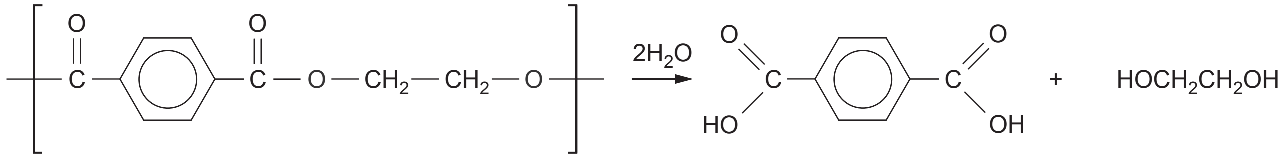
(iii) State the type of mechanism occurring in Stage 2. [1]

....

(iv) Draw the **skeletal** formula of the 5-carbon containing dicarboxylic acid used to produce Nylon 4,5. [1]

(d) The depolymerisation of polyamides and polyesters present a number of difficult problems as these polymers are very stable and only slowly decompose in the environment.
Poly(ethyleneterephthalate) (PET) is very difficult to hydrolyse but a new process using specific enzymes is proving promising. In this process 90% of PET is depolymerised into benzene-1,4-dicarboxylic acid.



Calculate the mass of benzene-1,4-dicarboxylic acid produced from 75 kg of PET if the yield from this hydrolysis is 90%. [3]

mass of benzene-1,4-dicarboxylic acid produced = ... kg

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